

The empirical recurrence for OEIS A303466, recovered from its tail

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Abstract

OEIS A303466 records “Empirical recurrence of order 93” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 142$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 95$. Its last coefficient is $c_{93} = 450868486864896 \neq 0$, so no further tail satisfies anything shorter; any order-93 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A303466 is “Number of $n \times 5$ 0..1 arrays with every element equal to 0, 1, 2, 3, 5 or 6 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

16, 384, 5237, 79377, 1243692, 19374638, 302023677, 4716032889, ...

The entry states:

Empirical recurrence of order 93 (see link above)

As of the “Last modified” line on the live entry (Apr 24 2018 10:41:31, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 142$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 142$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 93: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 93.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 284$ exact terms from $n = 2$ on, it returns order 93 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{93} c_i a(n-i),$$

c_1	21	c_{25}	146548295982268	c_{49}	−68003302928721488160	c_{73}	13293096951027
c_2	−85	c_{26}	1484767139038687	c_{50}	562010761637612611453	c_{74}	−9285189791194
c_3	586	c_{27}	−866242016013394	c_{51}	−1022674346638672127117	c_{75}	3822056471531
c_4	−11712	c_{28}	3074016446801191	c_{52}	1262573768562192440893	c_{76}	−1314853363555
c_5	43941	c_{29}	−11274930954134686	c_{53}	−2346285231691475211165	c_{77}	1130628655438
c_6	−118121	c_{30}	21233298619734152	c_{54}	3655954305450363008492	c_{78}	−292896449161
c_7	2237575	c_{31}	−43095706907345212	c_{55}	−4057830720360629118120	c_{79}	−420171419461
c_8	−8189919	c_{32}	−888335940575176	c_{56}	4456514058963856632164	c_{80}	2422996588023
c_9	11904634	c_{33}	−126135938372195379	c_{57}	−5177826797220495727740	c_{81}	−729313062820
c_{10}	−188551946	c_{34}	241246188422355994	c_{58}	5522017122794006551369	c_{82}	242885381744
c_{11}	700119842	c_{35}	700853412627201983	c_{59}	−4536944243335650854590	c_{83}	−173741687873
c_{12}	−898051880	c_{36}	135283593872018114	c_{60}	2557188087808335789523	c_{84}	262887886505
c_{13}	8497644109	c_{37}	−698313891765650703	c_{61}	−2297076381469535170293	c_{85}	−75231788737
c_{14}	−30371447551	c_{38}	−4994539488511001941	c_{62}	2190428681188858726750	c_{86}	−98039311304
c_{15}	47755012834	c_{39}	2178813739486091637	c_{63}	89500778351204367759	c_{87}	−11603736619
c_{16}	−232734851780	c_{40}	301485114092478794	c_{64}	−1301986704031566569274	c_{88}	97036744877
c_{17}	662882820806	c_{41}	18149545767559121220	c_{65}	1189476117513457483544	c_{89}	−25519843701
c_{18}	−1489122216787	c_{42}	−16788615072690113967	c_{66}	−2942619522745888965556	c_{90}	84824541955
c_{19}	4156274773873	c_{43}	12391356771445193640	c_{67}	7137874993867662967536	c_{91}	−8470311733
c_{20}	−5430776749945	c_{44}	−38867303549439405537	c_{68}	−11435540282434487534720	c_{92}	−8107218117
c_{21}	24467630051063	c_{45}	57151044078126337728	c_{69}	15219772785780903219408	c_{93}	4508684868
c_{22}	−45205992398065	c_{46}	−55905346549660393941	c_{70}	−17150619119771118207200		
c_{23}	−50126678161793	c_{47}	−14476951899256167296	c_{71}	15008946263024184548608		
c_{24}	−158548411526905	c_{48}	31734629353465742371	c_{72}	−13658600879049382382464		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 95$, and it is the recurrence OEIS A303466 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{93} - c_1x^{93-1} - \dots - c_{93}$. Its constant term is $-c_{93} = -450868486864896$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 93 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 93, so $\deg q = 93$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 93, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 93, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 450868486864896.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-93 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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